

trace_id	sol	timestamp_s	pos_x_m	pos_y_m	along_track_m	twt_ns	amplitude_dB	permittivity	ice_thickness_m	tid_krad	quality_flag
TR-0001	1	94.4	93.8	-7.1	94	21647.3	-41.62	3.0624	1855.513151	0.542	NOMINAL
TR-0002	1	177.5	185.5	-3.2	185.8	169839.5	-45.53	3.0961	14478.46338	0.577	NOMINAL
TR-0003	1	284	296.4	11.4	297.7	164535.9	-25.34	3.1901	13818.14615	0.735	NOMINAL
TR-0004	1	397.3	382.9	7	384.3	131950.8	-31.93	3.3418	10827.12272	0.817	NOISE_FLOOR
TR-0005	1	502.7	484.3	-0.6	486	95357.8	-49.2	3.4627	7686.702069	0.999	NOMINAL
TR-0006	1	591.5	577.3	7.4	579.3	23845.1	-27.13	3.4025	1939.060414	1.09	NOMINAL
TR-0007	1	674.2	693.8	9.4	695.8	140352.9	-51.49	3.2496	11678.78589	1.258	NOMINAL
TR-0008	1	779.9	779.5	-1.4	782.2	145118.1	-38.44	3.3735	11851.4772	1.333	NOMINAL
TR-0009	1	883.2	874	13.5	877.9	38346.4	-40.26	3.3779	3129.626158	1.496	NOMINAL
TR-0010	1	969.3	960.4	18.9	964.5	118968	-44.61	3.2979	9826.586429	1.577	CLUTTER
TR-0011	1	1059.4	1062.5	32.2	1067.4	133729.9	-55.42	3.4424	10811.57866	1.725	NOMINAL
TR-0012	1	1170.1	1156.1	26	1161.3	41835.8	-59.87	3.361	3422.98567	1.842	NOMINAL
TR-0013	1	1289	1266.6	26.3	1271.7	32728.3	-34.99	3.4208	2654.305242	1.941	NOMINAL
TR-0014	1	1377	1361.5	16.1	1367.2	181816	-23.1	3.4592	14663.42836	2.07	NOMINAL
TR-0015	1	1476.5	1446	12	1451.8	187416.6	-27.73	3.1197	15916.32863	2.154	NOMINAL
TR-0016	1	1579.2	1529.2	18.9	1535.2	157620.1	-20.88	3.2664	13081.82016	2.263	CLUTTER
TR-0017	1	1685.6	1647	8.9	1653.5	106870.6	-35.74	3.4822	8590.583255	2.463	CLUTTER
TR-0018	1	1795.8	1754.6	15.3	1761.3	84222.6	-33.13	3.1867	7077.001914	2.58	NOMINAL
TR-0019	1	1893.9	1844.6	2.2	1852.2	17702	-37.84	3.2942	1462.980758	2.611	NOMINAL
TR-0020	1	2002.2	1926.9	-10.8	1935.6	19528.7	-46.78	3.2571	1623.114086	2.758	NOMINAL
TR-0021	2	2101.6	2028.5	-4.1	2037.4	169299.4	-36.95	3.1215	14373.58192	2.897	NOMINAL
TR-0022	2	2197.9	2112.2	0.7	2121.3	76356.4	-43.56	3.4319	6182.574966	2.975	NOMINAL
TR-0023	2	2304	2218.1	-12.5	2227.9	142165.6	-27.98	3.0546	12201.3703	3.109	NOISE_FLOOR
TR-0024	2	2405.5	2303.7	-22	2314.1	95422.5	-25.02	3.0377	8212.390823	3.291	NOMINAL
TR-0025	2	2519.7	2387.6	-17.4	2398.1	109143.5	-59.41	3.0466	9379.537342	3.405	NOMINAL
TR-0026	2	2609.2	2483.9	-18	2494.4	166178.5	-23.9	3.0823	14198.04801	3.45	NOMINAL
TR-0027	2	2704.8	2600.9	-9.4	2611.7	64203.9	-32.14	3.3653	5249.776034	3.648	INVALID
TR-0028	2	2811.2	2700.4	-18.7	2711.6	52315.4	-57.66	3.3679	4276.034646	3.696	CLUTTER
TR-0029	2	2903.8	2782.4	-19.4	2793.6	175812.1	-38.75	3.0284	15154.21351	3.861	NOMINAL
TR-0030	2	3017.8	2865.2	-32.4	2877.4	165680.2	-43.85	3.4708	13339.73109	3.987	NOMINAL
TR-0031	2	3121	2946.7	-44.9	2959.9	129697.7	-37.39	3.1582	10947.2168	4.076	NOISE_FLOOR
TR-0032	2	3227.8	3039.3	-52	3052.8	37034.6	-34.18	3.2286	3091.660492	4.263	NOMINAL
TR-0033	2	3345.2	3119.7	-48.3	3133.2	113086.9	-56	3.2688	9382.301471	4.341	NOMINAL
TR-0034	2	3430.5	3213.6	-61.3	3228.1	56994.1	-48.6	3.2191	4764.901475	4.464	NOMINAL
TR-0035	2	3522.6	3333	-52	3347.8	107093.6	-33.29	3.2773	8873.53523	4.623	NOMINAL
TR-0036	2	3606.8	3448.1	-59.1	3463.1	170589.4	-30.3	3.0777	14585.79678	4.678	NOMINAL
TR-0037	2	3695.2	3541.8	-53.5	3557	164112.7	-39.78	3.1256	13924.08772	4.861	NOMINAL
TR-0038	2	3777.2	3647.2	-43.6	3662.8	21759.3	-46.66	3.0654	1864.200424	4.988	NOMINAL
TR-0039	2	3863.7	3744.9	-37.4	3760.7	112719.6	-55.53	3.4725	9073.38984	5.079	NOMINAL
TR-0040	2	3949.6	3826.3	-41.4	3842.2	111244.4	-42.81	3.0209	9600.663375	5.166	NOMINAL

trace_id	sol	timestamp_s	pos_x_m	pos_y_m	heading_deg	uncertainty_1sigma_m	loc_error_flag
TR-0001	1	117.3	118.9	-13.8	87.16	0.343	FALSE
TR-0002	1	224	213	-12	97.49	0.469	TRUE
TR-0003	1	334	330.1	-19.9	83.25	0.394	FALSE
TR-0004	1	421.1	426.6	-29.5	98.49	0.386	FALSE
TR-0005	1	517.5	533.4	-22.5	84.96	0.118	FALSE
TR-0006	1	625.6	628.7	-36.3	89.41	0.136	FALSE
TR-0007	1	742.3	722.6	-26.7	97.43	0.146	FALSE
TR-0008	1	848.7	818.6	-33.3	81.39	0.383	FALSE
TR-0009	1	942.8	919	-28	96.87	0.192	FALSE
TR-0010	1	1023.9	1034	-35.1	91.61	0.473	TRUE
TR-0011	1	1105.4	1137.9	-39.7	95.73	0.238	FALSE
TR-0012	1	1224.8	1222.5	-27.8	83.8	0.069	FALSE
TR-0013	1	1322.2	1323.3	-18.6	93.74	0.454	TRUE
TR-0014	1	1431.7	1411.2	-20.6	98.98	0.446	FALSE
TR-0015	1	1536.6	1517.7	-31.9	98	0.268	FALSE
TR-0016	1	1643.3	1610.8	-26	91.09	0.132	FALSE
TR-0017	1	1749.9	1706	-18.5	83.48	0.295	FALSE
TR-0018	1	1846.1	1819.3	-24.4	84.2	0.388	FALSE
TR-0019	1	1950.4	1912.2	-26.2	93.51	0.27	FALSE
TR-0020	1	2062.2	2030.6	-19.1	93.18	0.172	FALSE
TR-0021	2	2168.7	2135.4	-31.3	99.04	0.151	FALSE
TR-0022	2	2261.1	2247.6	-41.9	80.92	0.473	TRUE
TR-0023	2	2365.6	2358.4	-43.2	97.72	0.298	FALSE
TR-0024	2	2474.3	2453.7	-46.2	82.95	0.346	FALSE
TR-0025	2	2590	2568.1	-34.6	95.57	0.144	FALSE
TR-0026	2	2702.2	2676	-35.7	91.15	0.445	FALSE
TR-0027	2	2787	2761.3	-36.8	90.62	0.29	FALSE
TR-0028	2	2879.7	2871.5	-38.5	96.3	0.434	FALSE
TR-0029	2	2976.8	2987.8	-40.1	83.18	0.42	FALSE
TR-0030	2	3074.8	3097.9	-29.9	85.55	0.384	FALSE
TR-0031	2	3174.2	3187.4	-31.7	94.27	0.151	FALSE
TR-0032	2	3267.6	3303.2	-44.3	83.02	0.215	FALSE
TR-0033	2	3353.7	3391.7	-46.9	86.62	0.25	FALSE
TR-0034	2	3436.2	3505	-50.2	95.4	0.457	TRUE
TR-0035	2	3517	3620.3	-47.9	89.54	0.455	TRUE
TR-0036	2	3608.9	3715.9	-36.1	96.71	0.281	FALSE
TR-0037	2	3718.3	3827.8	-24.2	89.76	0.167	FALSE
TR-0038	2	3817.8	3923.4	-19.1	95.96	0.361	FALSE
TR-0039	2	3931.4	4040.2	-4.7	90.68	0.44	FALSE
TR-0040	2	4035.1	4146.7	-17.2	88.57	0.423	FALSE

trace_id	sol	slope_deg	roughness_index	surface_temp_K	tid_accum_krad	radar_noise_floor_dB	link_packet_loss_pct
TR-0001	1	4.54	0.308	91.5	0.145	-71.56	35.2
TR-0002	1	6.96	0.861	93.4	0.328	-68.24	30
TR-0003	1	1.3	0.862	89.5	0.435	-56.07	1.6
TR-0004	1	0.78	0.298	101.8	0.597	-74.04	18.9
TR-0005	1	2.86	0.858	94.3	0.723	-70.78	24.2
TR-0006	1	24.09	0.675	100.6	0.839	-72.22	23.1
TR-0007	1	23.22	0.561	103.2	0.977	-56.46	7.9
TR-0008	1	14.8	0.764	89.9	1.147	-72.46	27.4
TR-0009	1	20.14	0.737	96.3	1.251	-66.68	26.5
TR-0010	1	1.72	0.618	85.3	1.393	-67.74	17.3
TR-0011	1	9.05	0.813	94.2	1.508	-75.6	20.9
TR-0012	1	16.26	0.593	103.4	1.685	-68.38	33
TR-0013	1	14.81	0.358	89.9	1.878	-73.03	18
TR-0014	1	18.76	0.556	98.4	2.012	-67.64	13
TR-0015	1	12.19	0.384	90.2	2.14	-65.1	35.2
TR-0016	1	13.89	0.513	88.8	2.264	-69.84	22.2
TR-0017	1	6.01	0.481	99.2	2.409	-79.57	11.8
TR-0018	1	10.11	0.295	98.3	2.546	-66.16	13.8
TR-0019	1	24.92	0.54	92	2.691	-72.33	9.2
TR-0020	1	18.03	0.352	99.9	2.886	-62.75	7.7
TR-0021	2	18.47	0.321	96.8	3.062	-65.08	39.2
TR-0022	2	20.82	0.337	92.2	3.192	-62.3	5.1
TR-0023	2	1.14	0.144	90.8	3.386	-64.05	30.1
TR-0024	2	2.56	0.11	90.7	3.534	-71.48	38.6
TR-0025	2	6.31	0.789	87.3	3.641	-67.71	23.1
TR-0026	2	17.16	0.221	101.2	3.836	-77.87	9.9
TR-0027	2	13.95	0.433	96.9	3.998	-60.61	15.2
TR-0028	2	22.71	0.338	96.8	4.141	-65.78	2.4
TR-0029	2	24	0.179	100.3	4.304	-73.38	3.2
TR-0030	2	6	0.542	88.1	4.445	-62.77	18.8
TR-0031	2	0.82	0.33	90.7	4.631	-78.22	9.3
TR-0032	2	6.61	0.733	96.8	4.811	-75.05	4.6
TR-0033	2	5.62	0.219	90.3	4.925	-78.51	31.7
TR-0034	2	14.88	0.759	103.4	5.069	-68.28	12.2
TR-0035	2	10.06	0.318	95.8	5.213	-65.05	35.6
TR-0036	2	18.36	0.583	85.5	5.336	-63.14	34.4
TR-0037	2	14.68	0.117	100.3	5.518	-65.59	30.5
TR-0038	2	4.38	0.515	93.8	5.646	-55.08	17.4
TR-0039	2	20.36	0.875	94.4	5.787	-71.97	4.2
TR-0040	2	4.02	0.429	94.9	5.953	-69.99	30.4

sol	traverse_dist_m	valid_traces	bad_traces	min_ice_thickness_m	candidate_sites	mean_packet_loss_pct	mean_consensus_time_min
1	1841.6	16	4	7754.343608	4	19.845	5.969
2	1804.8	16	4	9750.038404	1	19.795	5.959

column	units	notes
trace_id	—	unique trace identifier
sol	sol index	sol number since landing
timestamp_s	s	elapsed mission seconds
pos_x_m	m	local ENU east coordinate
pos_y_m	m	local ENU north coordinate
along_track_m	m	cumulative along-track distance
twt_ns	ns	two-way travel time to basal reflector
amplitude_dB	dB	received signal amplitude at basal reflector
permittivity	—	bulk ice dielectric permittivity; pure ice ≈ 3.15
ice_thickness_m	m	derived: $d = (c \times \text{tw}_t_ns \times 1e-9) / (2 \times \text{sqrt}(\text{eps}_r))$
tid_krad	krad	total ionizing dose at radar electronics
quality_flag	enum	NOMINAL / CLUTTER / NOISE_FLOOR / INVALID
heading_deg	deg	clockwise from local North
uncertainty_1sigma_m	m	1 σ position uncertainty from VO
loc_error_flag	Boolean	TRUE when uncertainty exceeds ± 0.5 m (REQ.02.01)
slope_deg	deg	surface slope at measurement site
roughness_index	0–1	>0.7 flags physically inaccessible terrain
surface_temp_K	K	surface temperature at measurement site
tid_accum_krad	krad	accumulated TID at time of measurement
radar_noise_floor_dB	dB	radar receiver noise floor
link_packet_loss_pct	%	>30% triggers REQ.04.01 degraded-mode consensus
traverse_dist_m	m	total traverse distance in sol
valid_traces	count	NOMINAL quality traces only
bad_traces	count	CLUTTER + NOISE_FLOOR + INVALID
min_ice_thickness_m	m	mean thickness of NOMINAL traces (proxy for minimum)
candidate_sites	count	NOMINAL + thickness<5000m + slope<20° + roughness<0.7
mean_packet_loss_pct	%	mean inter-agent link packet loss for sol
mean_consensus_time_min	min	estimated consensus convergence: 2 + 0.2×packet_loss

field	value
mission_name	Europa Multi-Agent Ice Characterization Mission
mission_objective	Autonomously characterize Europa ice shell thickness and identify an optimal cryobot insertion site without real-time Earth support
destination	Europa surface
primary_instrument	Ground-penetrating radar sounder (analogous to Europa Clipper REASON)
thickness_accuracy_goal	±10% for ice depths 1–30 km (REQ.03.01)
along_track_resolution_m	≤100 m
localization_goal_m	±0.5 m per 50 m traverse segment (REQ.02.01)
autonomy_requirement	Fully autonomous; no real-time Earth uplink (REQ.01.01)
radiation_design_dose	≥100 krad TID (REQ.01.02)
consensus_degraded_goal	<10 min convergence at 30% packet loss (REQ.04.01)
version	v1.0 (sample data)